

CYBERSECURITY USING MACHINE LEARNING AND DEEP LEARNING

SEMINAR REPORT

Submitted in partial fulfilment with requirements of the award of degree

of

Bachelor of Technology

in

INFORMATION TECHNOLOGY

at

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY



Submitted by:

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COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

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CERTIFICATE

This is to certify that the seminar entitled “**CYBERSECURITY USING MACHINE LEARNING AND DEEP LEARNING**” is a bona-fide report of seminar is done by **SUDHANSHU SHEKHAR (14170234)** towards the partial fulfillment of the requirement for the degree of B. Tech in INFORMATION TECHNOLOGY at Cochin University of Science and Technology.

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ACKNOWLEDGEMENT

Many people have contributed to the success of this seminar. Although a single sentence hardly suffices, I would like to thank Almighty God for blessing me with his grace and taking my endeavor to a successful culmination.

I'm extremely grateful to our principal, **Prof. SUNIL KUMAR N** for the facilities provided by him during the preparation of this report.

I'm extremely thankful to the Head of the Department - Information Technology **Mrs. JAYAPRABHA P**, for giving me all the support and valuable directions in overcoming the difficulties faced during the preparation of this report.

I express my sincere thanks to my seminar coordinators **Mrs. SREELAKSHMY H** (Assistant Professor) and **Mrs. LISHA L** (Assistant Professor) Information Technology department for giving me innovative suggestions, timely advice, correction advices and suggestions during this endeavor.

I also express my gratitude towards all staff members of Information Technology department for their valuable advice and timely suggestions. I would also like to thank the faculties of Cochin University College of Engineering Kuttanadu (CUCEK) for their encouragement.

I would also express my deep sense of thanks to all the staffs and my classmates for their support, constructive criticism and suggestions.

ABSTRACT

Machine learning is adopted in a wide range of domains where it shows its superiority over traditional rule-based algorithms. These methods are being integrated in cyber detection systems with the goal of supporting or even replacing the first level of security analysts. Although the complete automation of detection and analysis is an enticing goal, the efficacy of machine learning in cyber security must be evaluated with the due diligence. We present an analysis, addressed to security specialists, of machine learning techniques applied to the detection of intrusion, malware, and spam. The goal is twofold: to assess the current maturity of these solutions and to identify their main limitations that prevent an immediate adoption of machine learning cyber detection schemes. Our conclusions are based on an extensive review of the literature as well as on experiments performed on real enterprise systems and network traffic.

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1. INTRODUCTION

The appeal and pervasiveness of machine learning (ML) is growing. Existing methods are being improved, and their ability to understand and answer real issues is highly appreciated. These achievements have led to the adoption of machine learning in several domains, such as computer vision, medical analysis, gaming and social media marketing. In some scenarios, machine learning techniques represent the best choice over traditional rule-based algorithms and even human operators. This trend is also affecting the cyber security field where some detection systems are being upgraded with ML components. Although devising a completely automated cyber defence system is yet a distant objective, first level operators in Network and Security Operation Centres (NOC and SOC) may benefit from detection and analysis tools based on machine learning. This paper is specifically addressed to security operators and aims to assess the current maturity of these solutions, to identify their main limitations and to highlight some room for improvement.

Cybersecurity is a set of technologies and processes designed to protect computers, networks, programs and data from attacks and unauthorized access, alteration, or destruction. A network security system consists of a network security system and a computer security system. Each of these systems includes firewalls, antivirus software, and intrusion detection systems (IDS). IDSs help discover, determine and identify unauthorized system behaviour such as use, copying, modification and destruction [2]. Security breaches include external intrusions and internal intrusions. There are three main types of network analysis for IDSs: misuse-based, also known as signature-based, anomaly-based, and hybrid. Misuse-based detection techniques aim to detect known attacks by using the signatures of these attacks. They are used for known types of attacks without generating a large number of false alarms. However, administrators often must manually update the database rules and signatures. New (zero-day) attacks cannot be detected based on misused technologies

Anomaly-based techniques study the normal network and system behaviour and identify anomalies as deviations from normal behaviour. They are appealing because of their capacity to detect zero-day attacks. Another advantage is that the profiles of normal activity are customized for every system, application, or network, therefore making it difficult for attackers to know which activities they can perform undetected. Additionally, the data on which anomaly-based techniques alert (novel attacks) can be used to define the

signatures for misuse detectors. The main disadvantage of anomaly-based techniques is the potential for high false alarm rates because previously unseen system behaviours can be categorized as anomalies.

Hybrid detection combines misuse and anomaly detection. It is used to increase the detection rate of known intrusions and to reduce the false positive rate of unknown attacks. Most ML/DL methods are hybrids.

Our study is based on an extensive review of the literature and on original experiments performed on real, large enterprises and network traffic. Other academic papers compare ML solutions for cyber security by considering one specific application and are typically oriented to Artificial Intelligence (AI) experts rather than to security operators. In the evaluation, we exclude the commercial products based on machine learning (or on the abused AI term) because vendors do not reveal their algorithms and tend to overlook issues and limitations.

1.1 CYBERSECURITY

Cyber security refers to the body of technologies, processes, and practices designed to protect networks, devices, programs, and data from attack, damage, or unauthorized access. Cyber security may also be referred to as information technology security.

1.1.1 IMPORTANCE OF CYBERSECURITY

It is important because government, military, corporate, financial, and medical organizations collect, process, and store unprecedented amounts of data on computers and other devices. A significant portion of that data can be sensitive information, whether that be intellectual property, financial data, personal information, or other types of data for which unauthorized access or exposure could have negative consequences. Organizations transmit sensitive data across networks and to other devices in the course of doing businesses, and cyber security describes the discipline dedicated to protecting that information and the systems used to process or store it.

1.1.2 CHALLENGES OF CYBERSECURITY

- a. Network security
- b. Application security
- c. Endpoint security
- d. Data security
- e. Identity management
- f. Database and infrastructure security
- g. Cloud security
- h. Mobile security

1.2 INTRUSION DETECTION SYSTEM

Intrusion detection systems help discover, determine and identify unauthorized system behavior such as use, copying, modification and destruction.

Security breaches include external intrusions and internal intrusions. There are three main types of network analysis for IDSs: misused-based, also known as signature based, anomaly-based and hybrid.

a. Misused-based, also known as signature based: Misused-based detection techniques aim to detect known attacks by using the signatures of the attacks. They are used for known types of attacks without generating a large number of false alarms. However, administrators often must manually update the database rules and signatures. New (zero-day) attacks cannot be detected based on misused technologies.

b. Anomaly-based: Anomaly-based techniques study the normal network and system behaviour and identify anomalies as deviations from normal behaviour. They are appealing because of their capacity to detect zero-day attacks. Another advantage is that

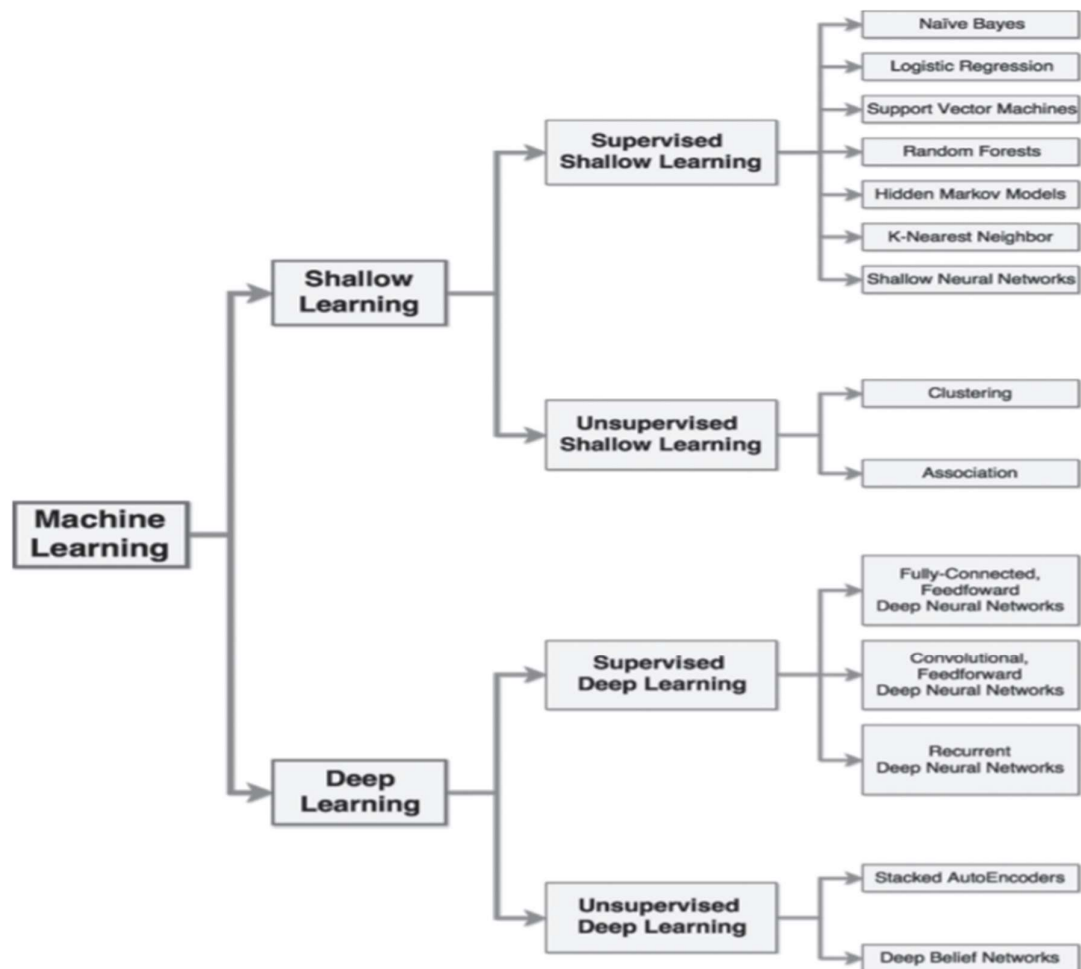
the profiles of normal activity are customized for every system, application, or network, therefore making it difficult for attackers to know which activities they can perform undetected. Additionally, the data on which anomaly-based techniques alert (novel attacks) can be used to define the signatures for misuse detectors. The main disadvantage of anomaly-based techniques is the potential for high false alarm rates because previously unseen system behaviours can be categorized as anomalies.

c. Hybrid: Hybrid detection combines misuse and anomaly detection. It is used to increase the detection rate of known intrusions and to reduce the false positive rate of unknown attacks. Most ML/DL methods are hybrids.

1.3 MACHINE-LEARNING AND DEEP LEARNING:

1.3.1 MACHINE-LEARNING: Machine learning includes a large variety of paradigms in continuous evolution, presenting weak boundaries and cross relationships. Furthermore, different views and applications may lead to different classifications.

1.3.2 DEEP-LEARNING: Deep learning is part of a broader family of machine learning methods based on artificial neural networks with representation learning. Learning can be supervised, semi-supervised or unsupervised.



Shallow learning and Deep learning approaches can be further characterized by distinguishing between supervised and unsupervised algorithms. The former techniques require a training process with a large and representative set of data that have been previously classified by a human expert or through other means. The latter approaches do not require a prelabelled training dataset. Each category can include dozens of different techniques

a. Shallow learning:

1. Supervised SL algorithms

- K-Nearest Neighbour (KNN). KNN are used for classification and can be used for multi-class problems. However, both their training and test phase are computationally demanding as to classify each test sample, they compare it against all the training samples.

- Support Vector Machines (SVM). These are non-probabilistic classifiers that map data samples in a feature space with the goal of maximizing the distance between each category

of samples. They do not make any assumption on the input features, but they perform poorly in multi-class classifications. Hence, they should be used as binary classifiers. Their limited scalability might lead to long processing times.

b. Deep learning:

All DL algorithms are based on Deep Neural Networks (DNN), which are large neural networks organized in many layers capable of autonomous representation learning.

1. Supervised DL algorithms:

- Convolutional Feedforward Deep Neural Networks (CNN). They are a variant of DNN where each neuron receives its input only from a subset of neurons of the previous layer. This characteristic makes CNN effective at analysing spatial data, but their performance decreases when applied to nonspatial data

- Recurrent Deep Neural Networks (RNN). A variant of DNN whose neurons can send their output also to previous layers; this design makes them harder to train than FNN. They excel as sequence generators, especially their recent variant, the long short-term memory.

2. Unsupervised DL algorithms:

- Deep Belief Networks (DBN). They are modelled through a composition of Restricted Boltzmann Machines (RBM), a class of neural networks with no output layer. DBN can be successfully used for pre-training tasks because they excel in the function of feature extraction. They require a training phase, but with unlabelled datasets.

1.4 DIFFERENCE BETWEEN MACHINE LEARNING AND DEEP LEARNING:

The difference between ML and DL include the following:

1. Data dependencies:

- Deep learning algorithms do not perform as well when the data volumes are small, because deep learning algorithms require a large amount of data to understand the data perfectly.
- The traditional machine-learning algorithm uses the established rules, the performance will be better.

2. Hardware dependencies:

- The DL algorithm requires many matrix operations. The GPU is largely used to optimize matrix operations efficiently. Therefore, the GPU is the hardware necessary for the DL to work properly.
- DL relies more on high-performance machines with GPUs than do traditional machine-learning algorithms.

3. Feature processing:

- The performance of most ML algorithms depends upon the accuracy of the features extracted.
- DL reduces the effort of designing a feature extractor for each problem.

4. Problem-solving method:

- machine learning usually breaks down the problem into multiple sub-problems and solves the sub-problems, ultimately obtaining the final result.
- In contrast, deep learning advocates direct end-to-end problem solving.

5. Execution time:

- deep learning algorithm takes a long time to train.

- machine learning algorithm takes short test time.

1.5 Network security data set:

Data constitute the basis of computer network security research. The size of the dataset also affects the training effects of the ML and DL models. Computer network security data can usually be obtained in two ways: 1) directly and 2) using an existing public dataset.

- Direct access is the use of various means of direct collection of the required cyber data. This approach is highly targeted and suitable for collecting short-term or small amounts of data, but for long-term or large amounts of data, acquisition time and storage costs will escalate.
- The use of existing network security datasets can save data collection time and increase the efficiency of research by quickly obtaining the various data required for research.

1.5.1 Types of network security data set:

a. DARPA INTRUSION DETECTION DATA SETS:

The dataset provides a large amount of background traffic data and attack data. It can be downloaded directly from the website.

b. KDD CUP 99 DATASET:

The KDD Cup 99 dataset is one of the most widely used training sets; it is based on the DARPA 1998 dataset.

c. NSL-KDD DATASET:

- The NSL-KDD dataset is a new version of the KDD Cup 99 dataset. The NSL-KDD dataset improves some of the limitations of the KDD Cup 99 dataset.

- This model identifies features between intrusive and normal connections for building network intrusion detectors. In the NSL-KDD dataset, each instance has the characteristics of a type of network data.

d. ADFA DATASET:

The ADFA data set is a set of data sets of host level intrusion detection system, which is widely used in the testing of intrusion detection products.

2. EXISTING SYSTEM AND DISADVANTAGES

A network security system consists of a network security system and a computer security system i.e., firewalls, antivirus software.

2.1 FIREWALL:

A firewall is a network security device, either hardware or software-based, which monitors all incoming and outgoing traffic and based on a defined set of security rules it accepts, rejects or drops that specific traffic.

Accept: Allow the traffic
 Reject: block the traffic but reply with an “unreachable error”
 Drop: block the traffic with no reply.

2.1.1 How firewall works

Firewall match the network traffic against the rule set defined in its table. Once the rule is matched, associate action is applied to the network traffic. For example, Rules are defined as any employee from HR department cannot access the data from code server and at the same time another rule is defined like system administrator can access the data from both HR and technical department. Rules can be defined on the firewall based on the necessity and security policies of the organization. From the perspective of a server, network traffic can be either outgoing or incoming. Firewall maintains a distinct set of rules for both the cases. Mostly the outgoing traffic, originated from the server itself, allowed to pass. Still, setting a rule on outgoing traffic is always better in order to achieve more security and prevent unwanted communication. Incoming traffic is treated differently. Most traffic which reaches on the firewall is one of these three major Transport Layer protocols- TCP, UDP or ICMP. All these types have a source address and destination address. Also, TCP and UDP have port numbers. ICMP uses *type code* instead of port number which identifies purpose of that packet.

2.1.2 Generation of firewall

Firewalls can be categorized based on its generation.

1. **First Generation- Packet Filtering Firewall:** Packet filtering firewall is used

to control network access by monitoring outgoing and incoming packet and allowing them to pass or stop based on source and destination IP address, protocols and ports. It analyses traffic at the transport protocol layer (but mainly uses first 3 layers). Packet firewalls treat each packet in isolation. They have no ability to tell whether a packet is part of an existing stream of traffic. Only It can allow or deny the packets based on unique packet headers.

2. **Second Generation- Stateful Inspection Firewall:** Stateful firewalls (performs Stateful Packet Inspection) are able to determine the connection state of packet, unlike Packet filtering firewall, which makes it more efficient. It keeps track of the state of networks connection travelling across it, such as TCP streams. So, the filtering decisions would not only be based on defined rules, but also on packet's history in the state table.

3. **Third Generation- Application Layer Firewall:** Application layer firewall can inspect and filter the packets on any OSI layer, up to the application layer. It has the ability to block specific content, also recognize when certain application and protocols. In other words, Application layer firewalls are hosts that run proxy servers. A proxy firewall prevents the direct connection between either side of the firewall, each packet has to pass through the proxy. It can allow or block the traffic based on predefined rules.

2.2 ANTIVIRUS SYSTEM

Antivirus software is a class of program designed to prevent, detect and remove malware infections on individual computing devices, networks and IT systems.

Antivirus software, originally designed to detect and remove viruses from computers, can also protect against a wide variety of threats, including other types of malicious software, such as Trojan horses, worms, rootkits, spyware, adware, botnets and ransomware.

2.2.1 How antivirus software works:

a. Antivirus software usually performs these basic functions:

- b. Scanning directories or specific files for known malicious patterns indicating the presence of malicious software;
- c. Allowing users to schedule scans so they run automatically;
- d. Allowing users to initiate new scans at any time; and
- e. Removing any malicious software, it detects. Some antivirus software programs do this automatically in the background, while others notify users of infections and ask them if they want to clean the files.

DISADVANTAGES:

- Disadvantages of Using a Firewall:

- 1. Firewalls cannot protect a network or pc from viruses, Trojans, worms and spyware which spread through flash drives, portable hard disk and floppy etc.
- 2. They may restrict authorized users from accessing valuable services.
- 3. it cannot protect the network from attacks from the inside.

- DISADVANTAGES OF ANTIVIRUS:

- a) Limited Detection Techniques
- b) Slows Down PC or Network
- c) Doesn't Fully Protect

3. NEW SYSTEM AND ADVANTAGES

ML AND DL ALGORITHM FOR CYBERSECURITY:

This section is divided into two parts i.e.

a. Traditional machine-learning algorithms in network security

i). K-Nearestneighbor:

The kNN classifier is based on a distance function that measures the difference or similarity between two instances. The standard Euclidean distance $d(x, y)$ between two instances x and y is defined as:

$$d(x, y) = \sqrt{\sum_{k=1}^n (x_k - y_k)^2}$$

Assume that the design set for kNN classifier is U . The total number of samples in the design set is S . Let $C \& D \{C_1; C_2; : : : C_L\}$ are the L distinct class labels that are available in S . Let x be an input vector for which the class label must be predicted. Let y_k denote the k th vector in the design set S . The kNN algorithm is to find the k closest vectors in design set S to input vector x . Then the input vector x is classified to class C_j if the majority of the k closest vectors have their class as C_j

KNN are used for classification and can be used for multi-class problems. However, both their training and test phase are computationally demanding as to classify each test sample, they compare it against all the training samples.

ii). Decision Tree:

A decision tree is a tree structure in which each internal node represents a test on one property and each branch represents a test output, with each leaf node representing a category.

In machine learning, the decision tree is a predictive model; it represents a mapping between object attributes and object value.

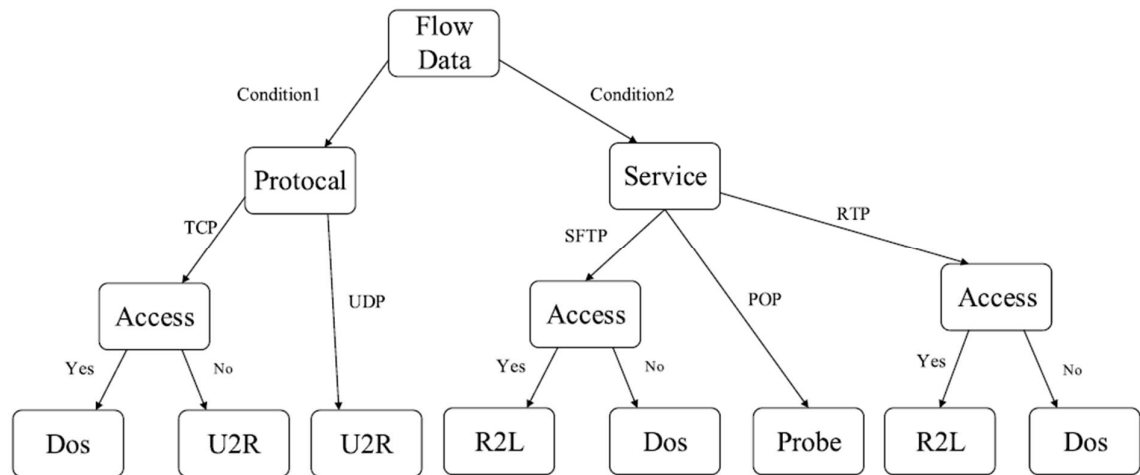


FIGURE 1. An example decision tree.

b. Deep learning in the field of cybersecurity

i). Deep belief network:

They are modelled through a composition of Restricted Boltzmann Machines (RBM), a class of neural networks with no output layer.

DBN can be successfully used for pre-training tasks because they excel in the function of feature extraction. They require a training phase, but with unlabeled datasets.

ii). Recurrent neural networks:

The recursive neural network (RNN) is used to process sequence data. In the traditional neural network model, data from the input layer to the hidden layer to the output layer; The layers are fully connected and there is no connection between the nodes between each layer.

The reason that RNN is a recurrent neural network is that the current output of a sequence is also related to the output before it.

iii). Convolutional neural networks:

Convolutional Neural Networks (CNN) is a type of artificial neural network that has become a hotspot in the field of speech analysis and image recognition.

Its weight-sharing network structure makes it more similar to a biological neural network, thus reducing the complexity of the network model and reducing the number of weights.

CNN is the first truly successful learning algorithm for training multi-layer network structures.

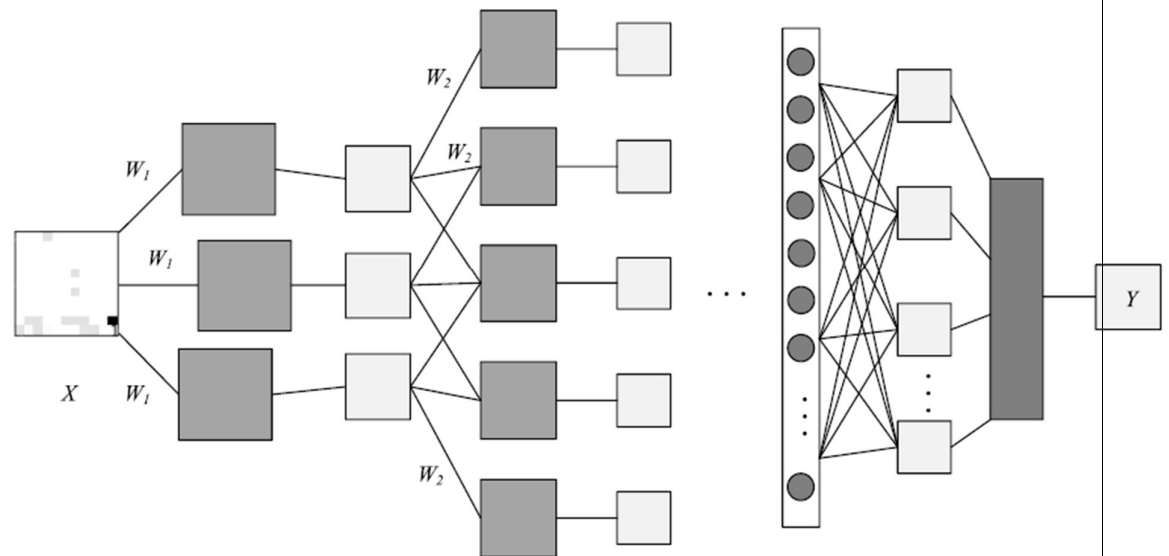


Fig. convolutional neural networks

4.DISADVANTAGES OF NEW SYSTEM:

- I. The ML and DL methods do not work without representative data, and obtaining such a dataset is difficult and time-consuming.
- II. There are many problems with the existing public dataset, such as uneven data, outdated content.
- III. Network information update very fast, which brings to the DL and ML model training and use with difficulty, model needs to be retrained long-term and quickly. So incremental learning and lifelong learning will be the focus in the study of this field in the future.

5. CONCLUSION.

Machine and deep learning approaches are increasingly employed for multiple applications and are being adopted also for cyber security, hence it is important to evaluate when and which category of algorithms can achieve adequate results. We analyse these techniques for three relevant cyber security problems: intrusion detection, malware analysis and spam detection. Then i explore several issues that influence the application of ML to cyber security. Our results provide evidence that present machine learning techniques are still affected by several shortcomings that reduce their effectiveness for cyber security. All approaches are vulnerable to adversarial attacks and require continuous re-training and careful parameter tuning that cannot be automatized. Moreover, especially when the same classifier is applied to identify different threats, the detection performance is unacceptably low; a possible mitigation can be achieved by using different ML classifiers for detecting specific threats. Deep learning is still at an early stage and no final conclusion can be drawn. Significant improvements may be expected, especially considering the recent and promising development of adversarial learning. Our takeaway is that machine learning techniques can support the security operator activities and automate some tasks, but pros and cons must be known. The autonomous capabilities of ML algorithms must not be overestimated, because the absence of human supervision can further facilitate skilled attackers to infiltrate, steal data, and even sabotage an enterprise.

6. REFERENCES

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